Promoting rice-upland crops systems to mitigate direct greenhouse gas emissions from intensive rice-based agriculture globally

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Promoting rice-upland crops systems to mitigate direct greenhouse gas emissions from intensive rice-based agriculture globally

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Abstract

Maximizing the rice-based cropping intensity supports food security but causes substantial methane (CH_4) and nitrous oxide (N_2O) emissions. Here, we project the spatiotemporal variations of their emissions from global rice-based ecosystems and assess the impacts of alternative cropping pattern pathways using a process-based model. Converting double-rice to rice-upland crops (DTRU) reduce global rice-based greenhouse gas emissions (GHGs) by ~28—32% relative to conventional (CONV) and by ~47—52% relative to shifts from single to double-rice (STDR) pathways. This implementation also dampens the emissions amplified by climate changes. Targeted expansion of rice-upland crop systems (TERU) partially restores rice production from DTRU with limited additional GHGs. Moreover, paired with carbon pricing, it yields lower increases in effective rice prices than other pathways. We quantitively demonstrates that selected structural shifts in rice-based cropping patterns would deliver robust and scalable GHG mitigations while preserving social welfare, offering a promising solution for global agricultural sustainability.

Key words: Rice, Greenhouse Gas, Cropping Patterns, Climate Change, Rice-upland crop systems

Introduction

Rice (*Oryza sativa L.*) is a main staple crop, with ~85% production concentrated in Asia, supporting over half of global population[1]. To cope with the increasing climate risks, rice-based agriculture would probably play a more important role in securing global food demand because the advantage of the relative short growing period (~ 80 – 170 days) allows high cropping intensities (i.e., number of crop cycles on the same land within one year), especially in subtropical and tropical monsoon regions[2,3]. However, growing evidences indicates significant declines in both rice cropping area and intensity in key double-rice cultivation regions due to climate, socioeconomic and policy factors, threatening food security[3-5]. Fully development of cropping intensity remains a fundamental approach to boosting production potential[6,7]. Therefore, leading rice producers and developing nations have launched administrative forces to improve and support muti-rice-cropping agriculture[4,8,9]. For example, China has promoted the recovery of double-season rice since the 2010s, while Vietnam and Indonesia have also encouraged expansion of multi-seasons rice cultivation[10-12].

Consequently, such intensifications could fuel more challenges to global climate mitigation strategies as rice is arguably the largest warming contributing food crop[13,14]. Rice paddies are among the largest anthropogenic sources of greenhouse gases (GHGs), particularly methane (CH_4) and nitrous oxide (N_2O) with global warming potentials (GWP) 14–34 and 256–298 times larger than that of CO_2, respectively[15,16]. Characterized by prolonged flooding, rice-paddy soils create strict anaerobic conditions that favor methanogenesis and inhibit aerobic CH_4 oxidation, acting major hotspots for CH_4 emissions[17,18]. Meanwhile, the extensively applied nitrogen fertilizer and frequent soil dry-wet cycles due to changing water regimes, such as midseason drainage and reflooding, stimulate nitrification and denitrification to produce and emit N_2O[19,20]. Overall, effective decisions for achieving low-emission rice-centered agriculture require a comprehensive understanding of its pattern both at the present and under future scenarios, yet existing quantifications remain incomplete and highly uncertain[18,21]. First, previous assessments often oversimplified rice-based ecosystems diversity, applying fixed rice-paddy maps that overlook seasonal rotations with upland crops such as widely employed rice-wheat and rice-rapeseed systems[22-25]. Second, most studies explicitly excluded CH_4 and N_2O fluxes during non-rice growing periods, leading to incomplete annual budgets[26-28]. Third, joint impacts of environmental (e.g., warming and elevated atmospheric CO_2) and management practices (e.g., N fertilization and water regimes) changes are poorly understood and rarely quantified at the global scale[14,16,29]. Finally, despite increasing recognition of agronomy adaptation approaches, few global studies have quantitatively evaluated the potential contributions of structural changes in rice-based cropping patterns in global climate mitigation[30].

Therefore, we aim to provide a comprehensive projection of future GHG emissions from global rice-based ecosystems using a process-based model, focusing on the role of structural shifts in different rice-based cropping patterns for sustainable development of rice agriculture (Fig.1). The specific objectives are to: (1) project the spatiotemporal variations of GHG emissions with current conventional (CONV) mixed rice cropping patterns; (2) assess the effects of single to double-rice (STDR) pathway and double-rice to rice-upland crop substitution (DTRU) pathways and their responses to changing climate; (3) evaluate the possible trade-offs of targeted expansion of rice-upland crop cultivation into upland areas (TERU) to reconcile production losses and GHG emissions; and (4) concisely asses the possible influences on world rice prices for economic and environmental balance. The study would lay the scientific foundation for applying structural changes to achieve sustainable development of global rice agriculture.

Results

Rice-based GHG emissions under conventional cropping pattern

. During the reference period (1995 – 2014), both CH_4 and N_2O emissions from global rice-based ecosystems increased significantly with means of 24.82 (±0.40) TgC yr^{-1} and 0.34 (±0.024) TgN yr^{-1}, respectively, corresponding to a total GWP of 992.56 (±17.10) TgCO2eq yr^{-1}. Projection showed a nonlinear trajectory of rice-related GHG emissions under current conventional cropping patterns which comprises single rice, double rice, and rice-upland crop systems across both SSPs. CH_4 and N_2O emissions are projected to increase initially before declining later in the century despite a continuously reduction in total physical area of rice-based ecosystems (Supplementary Fig.S4). Under SSP1-2.6, total GHG from global rice-based ecosystems are expected to increase by ~22% in the period of 2050–2075 (1202 $TgCO_2eq\ yr^{-1}$) relative to the reference period, peaking in the 2050s (~1233 $TgCO_2eq\ yr^{-1}$). However, projections under SSP5-8.5 scenarios suggest a decreased global rice GHG by 2100 (~3%, Fig.2ab) after a significant growing trend until the 2070s. The overall trends are driven primarily by reductions in CH_4 emissions (~85%), while N_2O emissions are projected to show smaller relative variations. Spatially, >64% of global rice cultivation regions are projected to experience increased GHGs under both SSPs (Fig.2c). Temperate single-rice cropping regions like Northeast China, Korea, and Japan would show more significant increases in GHGs under SSP5-8.5, whereas rice-based ecosystems located in subtropical and tropical climates present stronger GHG emissions under SSP1-2.6 (Supplementary Fig.S1). Notably, central China and southern India are projected to experience net reductions in rice-based GHG emissions.

For the underlying drivers of shaping GHG emissions trajectories, physical rice area dynamics had an essential role. Simulation experiments revealed the projected decreasing rice physical area associated land-use changes would cause >11% reduction in GHG emissions by2100 (Extended Data Fig. 2a, Supplementary Text 1). Trend decomposition further showed the such area loss substantially offset the influence of increases in weighted GHG fluxes, yielding net emission reductions toward the end of the century. Moreover, the extent of single-rice involved cultivation appears to be more influential for reducing total GHG under both SSPs, especially under the extreme scenario (Extended Data Fig. 2bc). Given the likely decline in future rice area, improving current cropping intensities is evitable for food supply. Thus, the importance of variations in different rice-based ecosystem area implies an opportunity to address the associated GHG challenges through structural changes of rice-based cropping patterns.

Mitigation effects of double-rice to rice-upland crop substitution (DTRU) pathway

There are two pathways for improving rice-based cropping intensity: (1) the intensification-based single to double-rice pathway (STDR), which promotes full recovery of double-season rice, and (2) the mitigation-focused DTRU pathway, which recommends rice-upland crops systems to replace double-rice systems. The former would increase rice-based GHG emissions by 43% and 35% under SSP1-2.6 and SSP5-8.5 compared with conventional pathway, largely driven by enhanced CH_4 emissions (Supplementary Fig.S7a). For instance, emissions from China's rice-based ecosystems rise by ≥2 Tg CH_4-C yr^{-1}, and Southeast Asia would present stronger sensitivity to this conversion, with CH_4 fluxes increasing by ~240% in Indonesia and ~120% Myanmar (Supplementary Fig.S7bc). Additional simulation experiments indicated that projected N management and irrigation changes would already decrease GHG emission globally for STDR, but the modest magnitudes of these effects

(<7%) underscore the need of structural efforts for mitigation (Supplementary Text 2, Fig.S8).

Our collected site-level paired crop yields revealed insignificant difference between energy yields of double rice and various rice-upland systems globally (Extended Data Fig.3). Therefore, focusing on the effect of GHG emissions exclusively, the DTRU pathway offers a compelling strategy for mitigation. Relative to STDR, DTRU conversion pathway would decrease >50% GHG emissions with ~38.5% reduction in global rice harvest area throughout the 21st century under SSP1-2.6 (Fig.3ac). Under SSP5-8.5, projection showed that 37.7% less harvest area would lead to a 46.6% mitigation in GHG (Fig.3bd). The overall mitigations stem from the lower CH_4 emissions because of fewer rice cultivation seasons and shorter flooding periods. However, the mitigation effects also exhibit spatial heterogeneity, as parts of northern Bangladesh and eastern India have a risk in presenting opposite responses due to the offsetting increases in N_2O emissions during non-rice growing seasons (Extended Data Fig. 5). In addition, national-level assessments reveal inconsistent effects across the globe. China would probably realize the largest mitigation benefit from implementing DTRU to replace STDR (>180 Tg CO_2e yr^{-1} reduction). Under both SSPs, China would contribute the largest share of global mitigation effect (>25%), followed by Indonesia (~12%), Myanmar (~10%), and India (~6%). Countries dominated by double-rice systems, especially in Southeast Asia under favorable environmental conditions, could achieve >60% reductions in annual rice-based GHG emissions with this pathway (Extended Data Fig.4).

Besides direct mitigation, the DTRU pathway confers climate-resilience co-benefits. With STDR pathway, concurrent variations in meteorological factors and elevated atmospheric CO_2 are predicted to increase both GHG emissions by ~4.2% under SSP5-8.5 until 2100. However, adopting DTRU transition would markedly alleviate such impacts by nearly half (Fig.4ab). Regarding SSP1-2.6, DTRU could even reverse the climate change effect on CH_4 emissions to negative. However, the consistent positive effect of climate changes on N_2O would be further amplified when implementing DTRU (Fig.4b). Specifically, the net climate-driven effects present great spatial variations. Under SSP1-2.6, the reductions in CH_4 emissions for double-rice systems in parts of Bangladesh and eastern India would be offset by increased N_2O emissions (Supplementary Fig.S9). For SSP5-8.5, the significant increased GHG emission from single-rice systems in the mid-high latitude regions (mostly CH_4 emission from northeast China) override the negative effects in subtropical regions under SSP5-8.5 (Fig.4cd). Notably, in major double-rice regions such as central China and eastern India, implementing DTRU pathway displays a convincing impact on alleviating the climate changes induced GHG, affirming a consistent mitigation potential of this strategy.

Potential of targeted expansion of rice-upland crops (TERU) conversion as compensation

To reconcile the reductions in rice cultivation area after adopting DTRU pathway, this study explored the responses of GHGs when expanding rice-upland crop systems by converting long-term upland croplands, termed TERU to prevent extreme shortage of rice. We designed four approaches each targeting different subsets of regions (Table S2).

Globally, transforming non-rice distributed irrigated upland croplands into rice-upland systems in climatically suitable region (apch3) probably delivers the most favorable trade-off: lowest increases in GHG emissions (~1.3% and ~2.7% for SSP1-2.6 and 5-8.5) relative to exclusive implementation of DTRU with comparable area gains (~2.6% SSP1-2.6 and ~6.3% SSP5-8.5) (Fig.5a). Although projected weighted GHG fluxes increase more sharply under SSP1-2.6, the larger climatically suitable area for multiple cropping causes more GHGs under SSP5-8.5. This approach is consistently attractive for India, which is projected to lead in absolute rice area gain (>1.8Mha) while incurring less additional CH_4 emissions across both scenarios (Fig.5b). Conversely, China and Myanmar would display the most sensitive responses of enhancing CH_4 emissions to TERU conversion, suggesting evitable climate risks (Supplementary Fig.S10). Interestingly, historically underutilized regions like southern USA and Brazil might emerge as potential suitable rice–upland zones. Projections show that TERU could be recommended in Brazil for the moderate enhanced CH_4 whereas such expansion in USA may less favorable.

Projected expenditure and rice price responses to cropping patterns shifts

As a key component of climate-policy, integrating a carbon price into world rice market would inevitably increase the equilibrium price despite changing rice-based cropping-patterns and SSP scenarios (Fig.6). Under CONV and STDR, the carbon tax on increased GHG emissions would ultimately offset the market supply effect from increment in rice harvested area (Extended Data Fig.6ab), causing at least 10% higher price relative to the reference period. In contrast, with DTRU pathway, substantial GHG reductions generate carbon credits that compensate >30% of the customer expenditure associated with area loss, yet rice price still rise by ~41% and ~63% under SSP1-2.6 and

SSP5-8.5, respectively. But with implementing TURE, rice price would become less expensive, whose net increases are often comparable or even less than those with STDR pathway. Such effects are evident with higher carbon prices and more stable carbon credits (Extended Data Fig.6cd).

Discussion

In the context of growing pressures on global GHG emissions and the need to stabilize food production under cropland loss in particular rice-paddies, our results underscore the importance of selecting appropriate rice-based cropping patterns while ensuring cropping intensity[6]. Given existing focus on agronomic adaptation strategies, including improved fertilizer and water management[28,31,32], our suggested structural adjustments can deliver a profound and durable GHG mitigation effect with limited side effects on economic welfare.

We quantitively demonstrate that implementing DTRU pathway, which replaces double-rice with rice-upland crop systems, would significantly decrease GHG emissions from global rice-based ecosystems this century. Nearly half of the reduced GHG compared with STDR pathway, that are currently promoted by major rice producers, substantially reconcile the apparent tension between food and climate as fulfilling the urgent needs of full developed cropping potential without affecting overall food energy production[8]. Plus, the robust mitigation potential of the DTRU pathway is further reinforced by its capacity to alleviate the positive feedback between GHG emissions from rice-based ecosystems and changing climate[14]. As the most climate sensitive crop species to emit GHG, such conversion introduces upland crops into systems, reducing the extent and frequency of flooding for rice cultivation[13,33]. This not only directly decreases CH_4 emission, which is highly sensitive to changing environmental conditions, but also disrupts soil anaerobic status, favoring aerobic microbial activities that suppress both CH_4 and N_2O pulses in subsequent cycles[34,35]. Meanwhile, increased N_2O emissions from introducing additional upland cropping seasons are less sensitive to climate change factors but are more responsive to management changes (e.g., N fertilization regimes), thus this implementation also helps to reduce the uncertainty of management mitigation efforts[36,37].

Despite the scalable mitigation potential, spatial variations in the effect of the DTRU transition emphasize a careful, regional-specific adaptive planning for such implementation. Specifically, leading rice producers, particularly in China, Indonesia, and Myanmar, would have substantial environmental benefits from this pathway consistently (Extended Data Fig. 4). However, under the background of projected reducing rice-cultivation area, India and Bangladesh may receive less pronounced impacts from this implementation, meanwhile less negative consequences of promoting maximum rice production pathway were noted in this region (Supplementary Fig.S7). Such divergent responses can be attributed to following reasons. First, rational rice-winter wheat systems already dominate northern India, the core agriculture region historically, because of the temperature limitation[38]. Second, rice CH_4 emissions during dry seasons which are designed to be substituted for DTRU are less intense than those from wet seasons (e.g., Kharif or Aman rice), the primary rice season in Southern Asia, in particular under rainfed conditions[39,40]. Plus, relative lower soil organic carbon (SOC) content shaped by local climates also contribute to the less sensitive response of CH_4 emissions to structural changes in rice-based cropping patterns[41,42]. Although significantly enhanced N_2O emissions from certain floodplain soils in southern Asia during non-rice growing periods may offset the effect of CH_4 reduction owing to high precipitation and lowland topography (Extended Data Fig.5)[20,43,44]. Such trade-offs would not override the overall mitigation effect on global scale due to the minor contribution of N_2O to total GHG budget[16,30].

To address concerns over potential rice production loss after implementing DTRU, we identified the least harmful TERU approach: transforming recent non-rice distributed irrigated upland croplands into rice-upland crop systems (apch3). This finding highlights the importance of land management history and irrigation instrument for sustainable crop relocating[45]. With interactive effects of sequential cropping, upland agriculture tend to have less SOC to fuel methanogenesis than continuous rice-existing fields, such as India the most TERU favorable country, which may have remarkable loss in rice cultivation area (Supplementary Fig.S10)[46]. Brazil appears to have great potential to contribute more rice-upland crop area with less environmental cost, but the conversion should strictly constrain to long-term cultivated soils rather than forested regions[47]. Furthermore, our selected approach3 emphasizes the establishment of irrigation, which enables flexible water regime as intermittent flooding to significantly mitigate rice GHG emissions globally[48]. Regarding the responses of N_2O emissions, integrating additional rice flooding seasons are likely to inhibit N_2O production than that of native upland crops (e.g., vegetables in southern China) given the massive N fertilization rates in these regions[49].

While TERU introduces severer GHG increments under high-emission scenarios, this trend stems from the increasing climatically suitable area (e.g., warming and wetting, Table S2) rather than direct intensification of climate induced GHG fluxes. In fact, the effects of changing climates may have been overrated by recent synthetic studies[16,50]. This discrepancy primarily arises from strong spatial variations in complex responses to key factors including warming and elevated CO_2, which are often exaggeratedly designed in short-term site-level experiments. For example, projected relatively modest warming in India with high ambient temperature constrains increase in CH_4 emissions[51,52]. Moreover, long-term significant enrichment in CO_2 (e.g., >10 years) reduces rice-derived GHG emissions due to stimulated methanotrophy and continuously enhanced mineral N uptakes, but are not fully captured in many short-duration measurements[50,53].

Climate mitigation actions are deeply intertwined with social welfare. The low-emission oriented sustainable agriculture ought to avoid excessive additional food expenditure[54]. Here, given the key role of placing carbon pricing in this century, we confirmed the superiority of co-implementing DTRU-TERU pathway from economic perspective. Rather than jeopardizing welfare for customers, our suggested structural transition pathway of rice-based ecosystems resolves the conflicts between economic burden and climate mitigation efforts. Adapting an effective mitigation management is the central to compensate the applied carbon tax. Moreover, these GHG mitigation achievements could become increasingly beneficial at higher carbon prices (e.g., \$100 tCO_2e^{-1}) especially under high-end climate change scenarios, underscoring the importance of coherent policy-management packages for both environmental effectiveness and economic acceptability[55]. Collectively, this study exhibits a strong and affordable GHG mitigation potential globally through shifting cropping pattern choices, especially DTRU-TERU also known as “double-rice to rice-upland cropping with targeted rice-upland crop expansion”. This pathway represents a systems-level transitions aligned with climate, food security and economic goals, offering a promising option for balancing the trade-off between welfare and GHG mitigation. Associated policy encouragements, social and technical tutoring, and market subsides would benefit promoting the recommended pathways, optimizing the overall positive effect on sustainability globally.

Several uncertainties and limitations remain. A key uncertainty source is the spatial distributions of rice-based ecosystems (Supplementary Fig.S12) and mixed rice-based cropping area, especially for obtaining a robust historical estimation[56]. Mapping the dynamics of mixed rice-upland cropping area over a large scale remains a crucial challenge[22,57] since cropping patterns are decided by individual farmers considering economic, environmental, and political factors, without a clear spatial variation patterns like climates[58,59]. Another important limitation is representation of different cultivars, which have shown crucial impacts on rice GHG emissions with great potential for future mitigation but hasn't been incorporated into earth system modeling[16]. Basically, higher yielding cultivars lead to reduced rice CH_4 emissions due to enhanced transport of O_2 from more porous root systems[60]. We used a linear relationship to roughly estimate the changes in rice plant productivity from advances in biotechnology (i.e., increase by 12.5% and 8.5% in the 21st century under SSP1-2.6 and SSP5-8.5, respectively, already considering compromise from climate change threats)[61]. However, recent biological improvement showed exciting achievements in low-CH_4 producing rice variety, implying another uncertainty source for the present study but an exciting progress toward low-emission agriculture[62]. In addition, influences of soil degradation and subsequent fallow activities should be explicitly integrated into future studies for evaluating the patterns and mitigation potential of GHGs from rice-cultivation globally[63,64]. Finally, the simplified modeled rice price changes are intentionally conservative and likely overrate rice price increases due to DTRU implementation (see Method), because of the substitutable nature as a cereal[65]. However, as a first order assessment, it sets up an upper boundary, implying the suggested structural transformation of rice-based systems is a promising opportunity for the sustainable social-environment balance. More sophisticated economic module including international trade and storage rate would be helpful to reduce the estimated uncertainties and more accurately support policymaking in future studies.

Methods

Process-based modeling of global rice-based GHG emissions and model performance evaluation

We used the TRIPLEX-GHG model v2.0, a process-based global ecosystem model which couples the major land surface processes (heat, energy, and hydrology cycles), plant phenology and physiology, vegetation dynamics, soil biogeochemistry (coupled C and N cycles), and varying management activities (agricultural practices and land use changes)[25,66]. The current study refined model representations of CH_4 and N_2O production and emissions from rice-based ecosystems. In short, the TRIPLEX-GHG v2.0 is capable of simulating the balance between methanogens (CH_4 production) and methanotrophy (CH_4 oxidation) to determine the net CH_4 fluxes over time under changing environment conditions (Fig.S1). Two CH_4 production pathways, acetoclastic and hydrogenotrophic methanogenesis are modeled explicitly to represent the characteristics of CH_4 sources under aerobic and anaerobic conditions, respectively. For CH4 oxidation, TRIPLEX-GHGv2.0 explicitly represented the activities of high and low affinity methanotrophy modulated by soil redox conditions. Regarding N_2O production, nitrification and denitrification are the predominantly responsible processes, with additional contributions from incorporated anaerobic ammonium oxidation (ANAMMOX) and nitrifier denitrification under flooding conditions. Three major gas transportation pathways are included in current model framework including diffusion, plant-mediate transportation, and ebullition for rice-based ecosystems. Only diffusion process dominates gaseous transportations under non-flooding seasons or circumstances. More detailed descriptions of key model processes are given in Supplementary Text 4.

Model performance was evaluated by comparing site-level observed flux data globally from published literatures (Supplementary Table S4). For each site, daily climate forcing data was obtained from the China Meteorological Forcing Dataset or CRU-JRAv2.5 (Climatic Research Unit (CRU) and Japanese reanalysis (JRA)). We employed three indexes D-value, RMSE, and pearson's *r* to exhibit good agreements when simulating the daily dynamics of CH_4 and N_2O fluxes (Supplementary Text 5, Fig.S2). Globally, CH_4 and N_2O fluxes from different types of rice-based ecosystems modeled by TRIPLEX-GHGv2.0 have a good consistency with measured results under different environmental and management conditions (closely aligned along the 1:1 line, Extended Data Fig. 1). Moreover, we compared our estimated historical global rice CH_4 emissions with previous modeled studies. Our estimation is within the range of existing estimates and closely align with the ensemble mean, reinforcing the model reliability (Supplementary Fig.S3).

Global model simulations and input datasets

Global simulations were conducted at a daily time step and at a 0.25 × 0.25° spatial resolution from 1840 – 2100 using a high-performance computer system (HPC). The model simulations underwent an initial 300-year spin-up to obtain a relative equilibrium in soil carbon and nitrogen pools driven by the multi-year average meteorological data before the analysis. Historical period simulation covered 1840–2014 while projections refer to the period of 2015–2100. Specifically, the projection simulations were conducted with two different Shared Socio-economic Pathways-Representative Concentration Pathways (SSP-RCP), SSP1-2.6 and SSP5-8.5, representing the changes in climate and management under the sustainable and the most extreme pathway, respectively.

For model inputs, five categories of datasets were needed, which are climate forcing data, atmospheric conditions data, initial soil and vegetation data and, field management data. Climate forcing data required global daily meteorological data, including maximum, minimum air temperature, mean surface temperature (K), precipitation (mm s^{-1}), specific humidity (g g^{-1}), and wind speed (m s^{-1}). Both historical (1950—2014) and projection period (2015—2100) climate data which have already undergone bias-correction were obtained from the NASA Earth Exchange Global Daily Downscaled Projections (NEX-GDDP-CMIP6) at 0.25° spatial resolutions[67]. Two participating Coupled Model Intercomparison Project Phase6 (CMIP6) Earth System Models (ESMs) were employed, with ACCESS-CM2 representing high-climate-sensitivity products, and NorESM2 providing low-climate-sensitivity projections[68]. The variations in atmospheric components were also addressed in detail for simulations. Global monthly gridded atmospheric NHx and NOy depositions (Inter-Sectoral Impact Model Intercomparison Project, ISIMIP3b), and atmospheric CO_2 concentration data[69] were used to represent the nonuniform distributed feature of these changes across the whole study period. As for atmospheric CH_4 and N_2O concentrations, we employed the annual global mean concentration data during simulations[70]. Regarding other environmental factors, Harmonized World Soil Database v2.0 delivered soil physical and chemical properties included soil sand, clay content, pH, soil organic carbon content (SOC), and C:N ratio of different layers for a 5m profile.

Field management inputs were more complicated with many aspects of agricultural decisions. Historical chemical N fertilizer and manure applications on grid level (1901—2019) were extracted from History of anthropogenic Nitrogen inputs (HaNi) dataset[71]. In terms of future part, gridded chemical N fertilizer application rates were based on LUH2 outputs under different SSPs (i.e., weighted mean by combining national fertilizer application rates in kgN ha^{-1} yr^{-1} per crop functional type and grid fraction of different crop function types and cropping intensity maps)[72,73]. Reversely, future manure application is projected to simply follow the same trends of annual synthetic N fertilizer application changes during 2015—2100 across different continents. The constructed projected global chemical N fertilizer and manure N usage align well with historical trends. This study also spatial explicitly define the timing of management practices, changes of water regimes and applications of fertilizer in particular. We integrated three dataset: RiceAtlas[2], ChinaRiceCalendar[74], and RICA[75] to ensure the most reliable, realistic, and finest spatial resolution of current utilized rice calendar data, including detailed global spatial rice crop calendar to determine the total number, planting, heading, and harvesting dates of rice-growing seasons. Previous studies often used fixed ratios on national level to determine the various water regimes of rice-paddies[76]. Instead, we assumed that for projection period, all irrigated rice fields are intermittent flooded with one mandatory mid-season drainage, whereas the subsequent variations and water regimes of rainfed rice fields are determined by water table dynamics submodule and an irrigation submodule which responds to meteorological variations. This allows a more realistic simulations under different climate and environmental conditions.

Different rice-based cropping patterns and simulation designs

Variations of the distribution of rice-based ecosystems are essential for the present study. Both historical and future land use information (e.g., upland croplands and rice-paddy fractions etc.) were obtained from Land-Use Harmonization 2 (LUH2)[72]. The historical results of physical area of rice are generally consistent with GloRicev1.0 dataset (Supplementary Fig.S12)[77]. Regarding the projection period, LUH2 data produced future distributions of different land use types based on the frame work of integrated assessment models, IMAGE3.0 and REMIND–MAgPIE, for SSP1-2.6 and SSP5-8.5, respectively. It is important to note the changing area and distribution of rice are the basis for all projections in this study because: 1) the high likely of loss in physical area for rice cultivation given the currently slowed increasing trends, environmental burden, and population burden; and 2) keep consistent with the management changes, which were produced from the same frameworks.

Notably, in different with rice harvest area, LUH2.0 provided rice physical area, indicating the presence of more than one season of rice cultivation in a grid cell. A major uncertainty source of current rice GHG budget is the mixed distribution of different rice cropping patterns (e.g., single rice, double rice, and rice-upland crops). Previous studies oversimplified this by directly using rice harvest area[18,32] or assigning double rice cropping to wherever appliable based on rice calendar products[78]. Subsequently, physical and harvest rice area are explicitly specified in this study by integrating GloRicev1.0 dataset. The ratio of rice harvest to physical area (RRHPA) reflects the fraction of single and double rice cropping of the pixel (i.e., =1.0 means all single rice and ≥2.0 means all double in the pixel). Regarding the distribution of rice-upland system, we employed the MIRCA2000 dependent cropping intensity maps to identify pixels which are multi-cropped but have single rice distributions[73].

Regarding projections, different pathways are designed to evaluate the responses of GHG emissions from rice-based ecosystems when implementing different pathways of cropping patterns shifts (Figure.1). First, we kept the RRHPA constant at the level of the 2010s for projections as the conventional pathway, which remain mixed rice-centered cropping patterns. Next, we tested the single to double-rice (STDR) pathway, which represents pursuing the maximum rice cultivation potential in the future by promoting double rice wherever applicable. This pathway represents the efforts that has been encouraged by major rice producing countries. In addition, we investigate the effects of double-rice to rice-upland crops (DTRU), which aimed to convert all double-rice cultivation area to rice-upland crops systems, and the subsequent upland crops were selected relied on the most dominant general crop types in the grid cell (e.g., C3/C4 annual crop, C3 N fixation crops or perennial crops etc.). This aligned well with current international needs in CH_4 reduction but affects rice harvest area, thus the key cereal product in food market. Therefore, finally, targeted expansion of rice-upland crops systems (TERU) was designed to converting a portion of existing suitable uplands to rice-upland cropping systems as a subside. Under projected changing climate, certain proportion of grids would be assigned with rice-upland systems expansion potential with following rules: 1) cropland already exist in the pixel; 2) annual growing degree days (GDD) are greater than 4800 ℃; 3) annual precipitations are greater than 1000mm[7]. Four strategies of TERU were designed to represent different recover levels of rice harvest area. Within the environmental favored region above, projected conversion to rice-upland crop systems area are also constrained by the distribution of irrigation instrument, elevation and slope as shown in Supplementary Table S2. For example, for approach3, 20% climate-favorable irrigated upland croplands in the grid cells without rice-paddy distribution for the last two years would be transformed into rice-upland crops systems.

Specifically, this study excluded ratoon rice cropping. Please refer to Supplementary Text 3 for greater detail.

Attribution of area, management, and climate changes impacts

This study applied simulation experiments approach to quantitively evaluate the effects of changing management practices and climate to GHG emissions through holding each factor consistent with the beginning of projection period of corresponding datasets (Supplementary Table S1). For example, by fixing climate change data at the beginning level of projection, the difference against multifactor modelled results would indicate the contributions of changing climate to total GHG emissions.

In addition, we applied the Logarithmic Mean Divisia Index (LMDI) (Eq.1—3) to attribute the trend of overall projected GHG emissions from rice-based ecosystems to changes in rice harvest area and weighted mean fluxes[79]. We explicitly separate the roles of double-rice and single-rice systems (incl. single rice and rice-upland crop systems) for both SSPs.

$$E(t) = A_d(t) \cdot F_d(t) + A_s(t) \cdot F_s(t) \qquad \text{Eq. 1}$$

$$\triangle E = E^t - E^{t-1} = w_d \cdot \ln\frac{A_d^t}{A_d^{t-1}} + w_d \cdot \ln\frac{F_d^t}{F_d^{t-1}} + w_s \cdot \ln\frac{A_s^t}{A_s^{t-1}} + w_s \cdot \ln\frac{F_s^t}{F_s^{t-1}} \qquad \text{Eq. 2}$$

$$w_x = \frac{E_x^t - E_x^{t-1}}{\ln E_x^t - \ln E_x^{t-1}} \ , \ E_d = A_d \cdot F_d,\ E_s = A_s \cdot F_s \qquad \text{Eq. 3}$$

Where A_d and A_s are the physical harvest area of double and single rice ecosystems, and F_d and F_s denote the corresponding mean annual GHG fluxes (g CO2eq m^{-2} yr^{-1}). $\triangle E$ is the year-to-year increment, which could be partitioned as four additive drivers, representing changes in double rice area, single rice area, fluxes from double rice and fluxes from single rice, respectively. w_x denotes the log-mean weight for each system ($x = d, s$). The LMDI was computed for two periods, 2015 – 2100 and 2075 – 2100, representing the whole projection period and the end of it.

A positive cumulative value denotes a net *increasing* effect on emissions, whereas a negative value denotes a *reducing* effect. And it is noteworthy that the contribution of changing area of different rice-based ecosystems do not imply the land use change effects, which are embedded in changing GHG fluxes component.

Energy yields comparison across different rice-based cropping systems

All crop productivity data were extracted from paired experiments at the same study sites from peer-reviewed and published articles (2000—2025) using Google Scholar. Our inclusion criteria were as follows: 1) only *in-situ* experiments were included; 2) each study reported yields for a double-rice (DR) system and at least one rice–upland system at the same site (paired design), which avoid inconsistent cropping density due to environmental heterogeneity; 3) the duration of the measurements was at least one year-round; and 4) ratoon rice was excluded at current stage. Ultimately, we retained 42 sets of mean crop yields at different site years for further analysis (Supplementary Table S3).

Here, we employed the total energy yields to evaluate the consequences of different cropping systems on food supply (Eq.4). Two-sided paired t-test on site-level annual energy yields were performed to show the difference. Mass yields are not commensurate across cereals, oilseeds, and tubers, whereas kilocalories per hectare per year (kcal ha^{-1} yr^{-1}) provide a common unit to assess a system's contribution to food energy supply and to the diversification of diets.

$$EY_i = \sum MassYield_i \cdot EnergyDensity_i \quad \text{Eq.4}$$

Where i represent different crop species and crop-specific energy densities (kcal 100 g^{-1} raw cereal) were obtained from Nutrient Optimiser mainly including: rice (353 kcal 100 g^{-1}), wheat (334 kcal 100 g^{-1}), maize (356 kcal 100 g^{-1}), rapeseed (668 kcal 100 g^{-1}), and potato (77 kcal 100 g^{-1}).

Rice price and expenditure responses to implementing different cropping patterns changes

In order to provide a first-order assessment of possible responses of rice price and consumer expenditure, key welfare indicators, to proposed alternative rice-cropping pathways, we employed a concise and straightforward global market-clearing model with constant elasticities of supply and demand. Hence, we also evaluated the effect after greenhouse-gas (GHG) changes are priced (carbon or emission pricing). We compared two periods: 1995—2014 (reference) and 2050—2075 (projection) across cropping pattern change pathways and SSPs, treating each period as a steady state.

This framework abstracts from international trade, storage, and inter-commodity competition, yielding a transparent mapping from exogenous supply shifts (via harvested area) and carbon pricing (tax or credit) to prices, demand quantities, and expenditures. The results provide conservative, decision-relevant bounds on welfare impacts attributable to cropping-pattern change and carbon pricing.

First, we compute the exogenous supply shock fraction σ as a result of the changes in harvested area (Eq.5). For each pathway, we translate harvested area changes into an exogenous leftward shift of the global rice supply curve at the baseline price, P_0. Therefore, it expressed as a fraction of baseline supply (Q_0, baseline world rice production ~ 475 million metric tons yr^{-1}, USDA) removed at the baseline price (P_0):

$$\sigma = -\frac{\Delta Q_S}{Q_0} \quad \text{Eq.5}$$

Where ΔQ_S denote the change in supply at P_0, determined by change in harvested area.

Second, based on the market equilibrium with constant-elasticity model, the demand $Q_D(P)$ and post-shock counterfactual supply $Q_S(P)$ curves are as follows:

$$Q_D(P) = Q_0 \cdot \left(\frac{P}{P_0}\right)^{-\varepsilon_D} \text{ Eq.6}$$

$$Q_S(P) = Q_0 \cdot (1-\sigma) \cdot \left(\frac{P}{P_0}\right)^{\varepsilon_S} \text{ Eq.7}$$

We used 164.8 (USD ton^{-1} free on board), a mean value from a previous study[80] as baseline world rice price P_0. ε_D is the Marshallian own-price elasticity of demand (in absolute value), ranging from 0.4 – 0.7 with 0.55 as optimal. This reflects inelastic staple demand in global aggregation, consistent with household studies of rice almost ideal demand system (AIDS) studies. ε_S means own-price elasticity of supply which are set to range 0.10 – 0.20 with 0.15 as optimal for the present study. Rice supply is unusually inelastic globally due to land/water constraints and policy insulation but in our study these changes are embedded. We adopt a conservative long-run range to avoid overstating price dampening and double counting of area responses.

Subsequently, the market clearing equilibrium $Q_D(P_1) = Q_S(P_1)$ solves the post-supply shock market price (P_1) and quantity (Q_1) of rice.

Therefore, the global consumer rice expenditure change (ΔE in USD yr^{-1}, >0 indicates a burden) is:

$$\Delta E = P_1 \cdot Q_1 - P_0 \cdot Q_0 \text{ Eq.8}$$

Meanwhile, this study integrated carbon credit (from GHG mitigation) and carbon tax (from GHG increase) into the formulation, named carbon revenue (R).

$$R = (GHG_{ref} - GHG_{proj}) \cdot carbon\_price \quad \text{Eq.9}$$

Finally, the effective rice price after returning carbon revenue as a per-unit rebate for consumption is computed as Eq.10. And the relative different with baseline price is the key indicator of welfare.

$$P_2 = P_1 - \frac{R}{Q_1 \cdot P_0} \quad \text{Eq.10}$$

GHG_{ref} *and* GHG_{proj} is the GHG changes in equivalent CO_2 during two periods. We had two uniform $carbon\_price$ levels, 50 and 100 USD tCO_2e^{-1}, representing the lower and upper boundary for achieving Paris Agreement[81].

Data availability

The data used in this study are publicly available. The daily climate forcing data for both historical and projection periods are from the NASA Earth Exchange Global Daily Downscaled Projections (NEX-GDDP-CMIP6) at https://www.nccs.nasa.gov/services/data-collections/land-based-products/nex-gddp-cmip6. Atmospheric N deposition data can be obtained from https://data.isimip.org/search/page/2/product/InputData/subcategory/n-deposition/tree/ISIMIP3b/. Soil physical and chemical property datasets are from Harmonized World Soils Database version 2.0 (https://gaez.fao.org/pages/hwsd). The concentration of atmospheric CH_4 and N_2O are obtained from https://doi.org/10.15138/P8XG-AA10 and https://doi.org/10.5194/gmd-13-3571-2020 for historical and projected periods, respectively.

Code availability

The source code and configuration for the current version of the TRIPLEX-GHG model v2.0 employed for the simulations is available at https://doi.org/10.5281/zenodo.7654741. Notably, please contact the corresponding author if you plan an application of the model and longer-term scientific collaboration.

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Figures

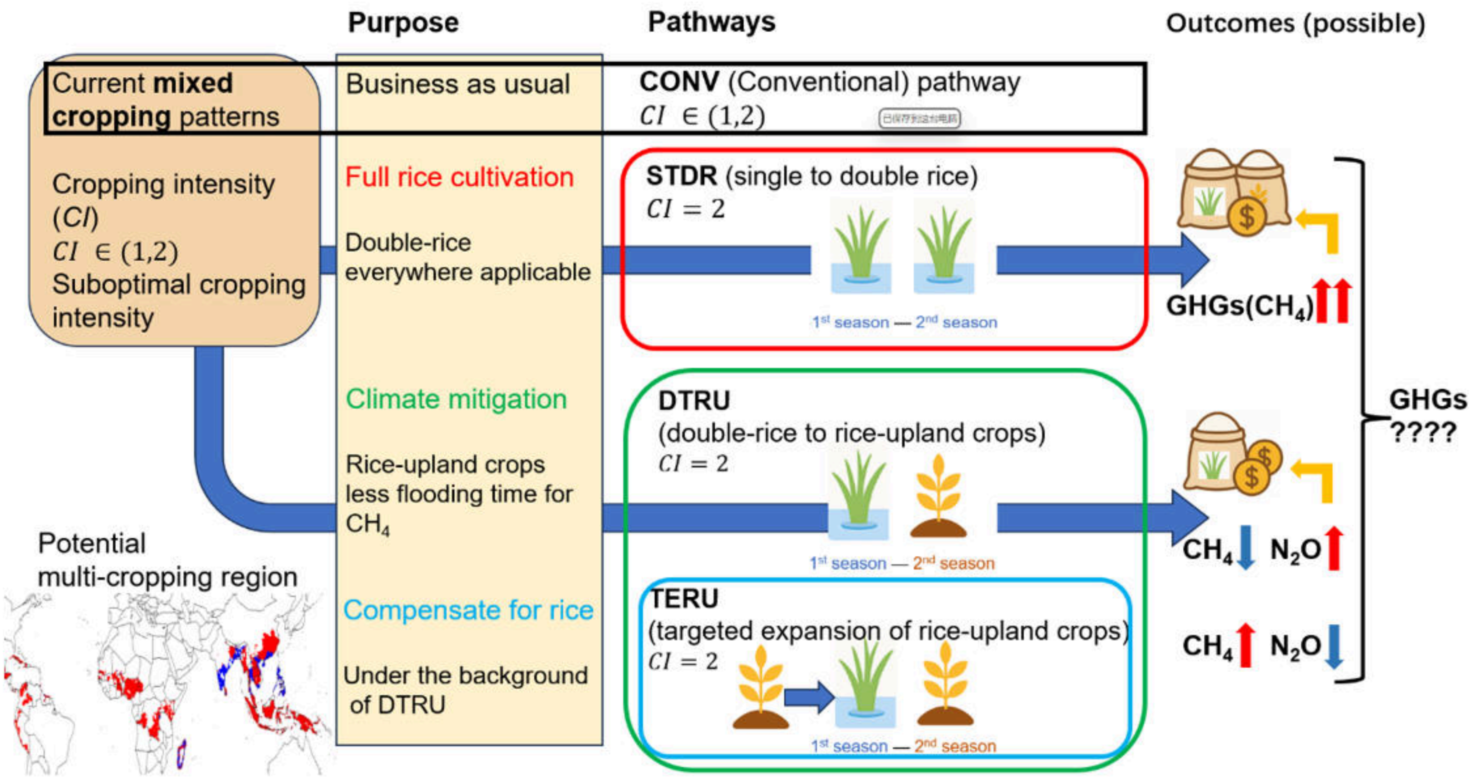

Figure 1

Schematic illustration of the framework of the study design. Current suboptimal cropping intensity of global rice-based ecosystems with the mixed rice-based cropping patterns ought to be improved for future food security. Therefore, STDR and DTRU pathways represent two possible trajectories with different purposes, while TERU acts as a possible supplement. The possible outcomes include responses of rice-price as a welfare indicator and overall greenhouse gas emissions. The map at bottom right shows inadequately cropped double-rice region (red) in the 2000s.

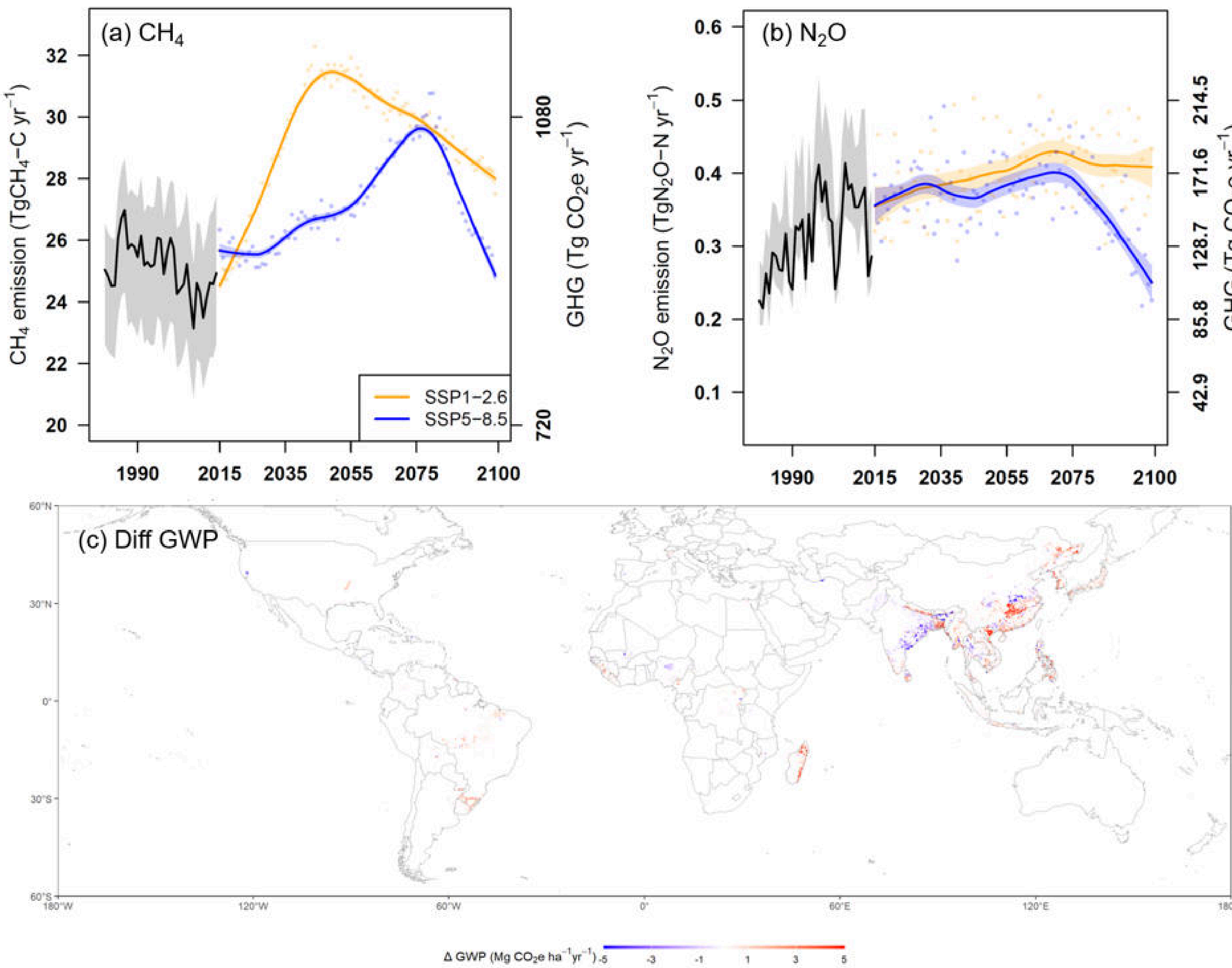

Figure 2

Spatiotemporal variation patterns of projected CH_4 and N_2O emissions from global rice-based ecosystems with conventional cropping pattern (CONV) under SSP1–2.6 and SSP5–8.5.

(a,b) projected temporal variations in CH_4 and N_2O emissions, respectively. Grey shading denotes uncertainty arising from alternative cropping-pattern assumptions; The dots show the mean annual emissions while the colored lines represent the LOESS smoothed emission trends and the corresponding shaded areas are the 90% confidence intervals (CI) of the fitted curves. (c) Global distribution of the mean change in GWP over the “neutrality period” (2050–2075) relative to the reference period (1995–2014), averaged across both SSPs.

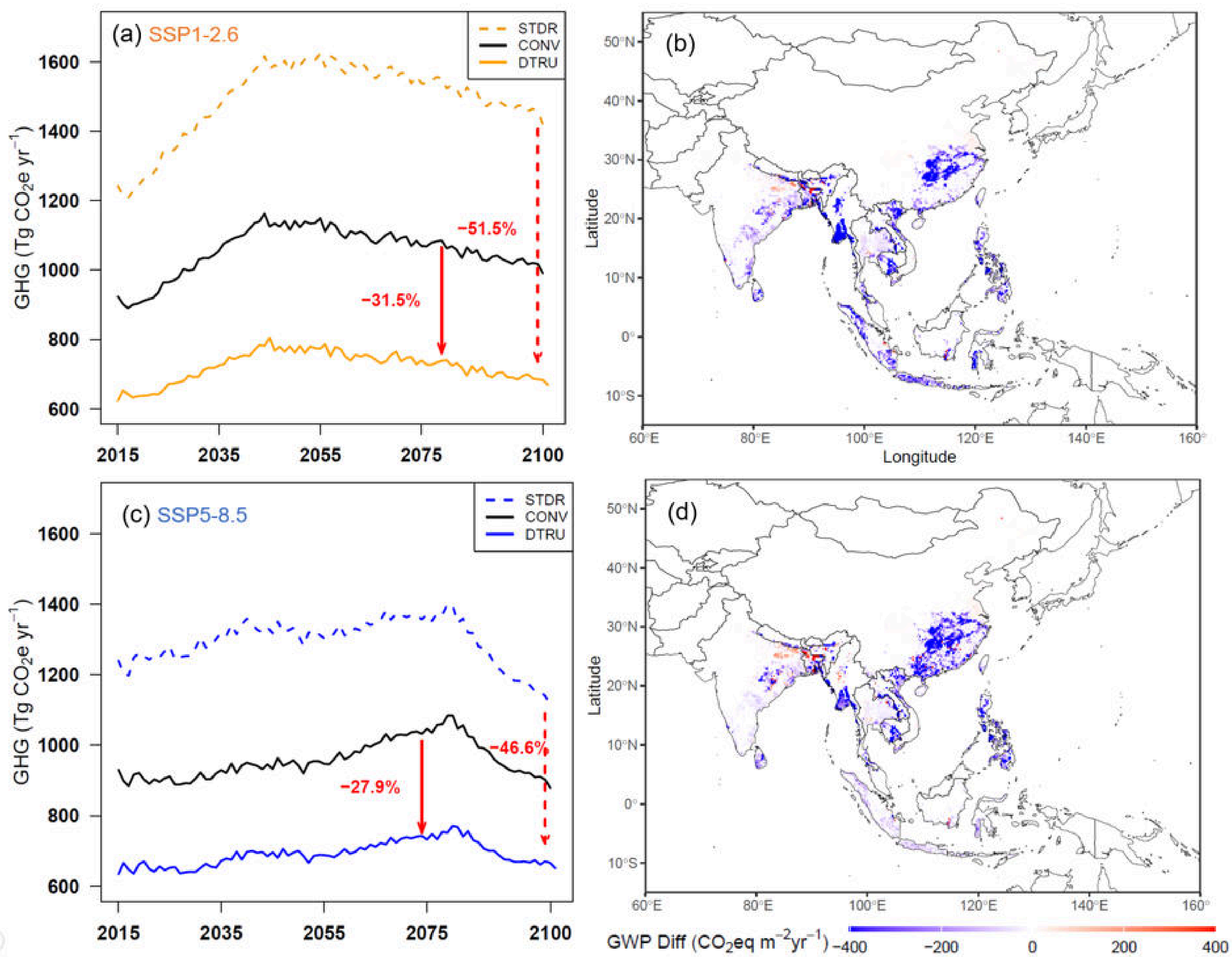

Figure 3

Projected impacts of the double rice to rice-upland crop system (DTRU) pathway on global rice-based ecosystem GHG emissions under SSP1–2.6 and SSP5–8.5.

(a,c) temporal variations of projected GHG emissions from rice-based ecosystems with conventional (CONV), single to double-rice (STDR) and double-rice to rice-upland crops (DTRU) patterns and their differences under SSP1-2.6 and SSP5-8.5, respectively. Panel b and d show spatial variations of the effects of shifting from STDR to DTRU pathway over 2050 – 2075 under SSP1-2.6 and SSP5-8.5, respectively. As the most important rice producing region, Asia is zoomed in for a clear visualization.

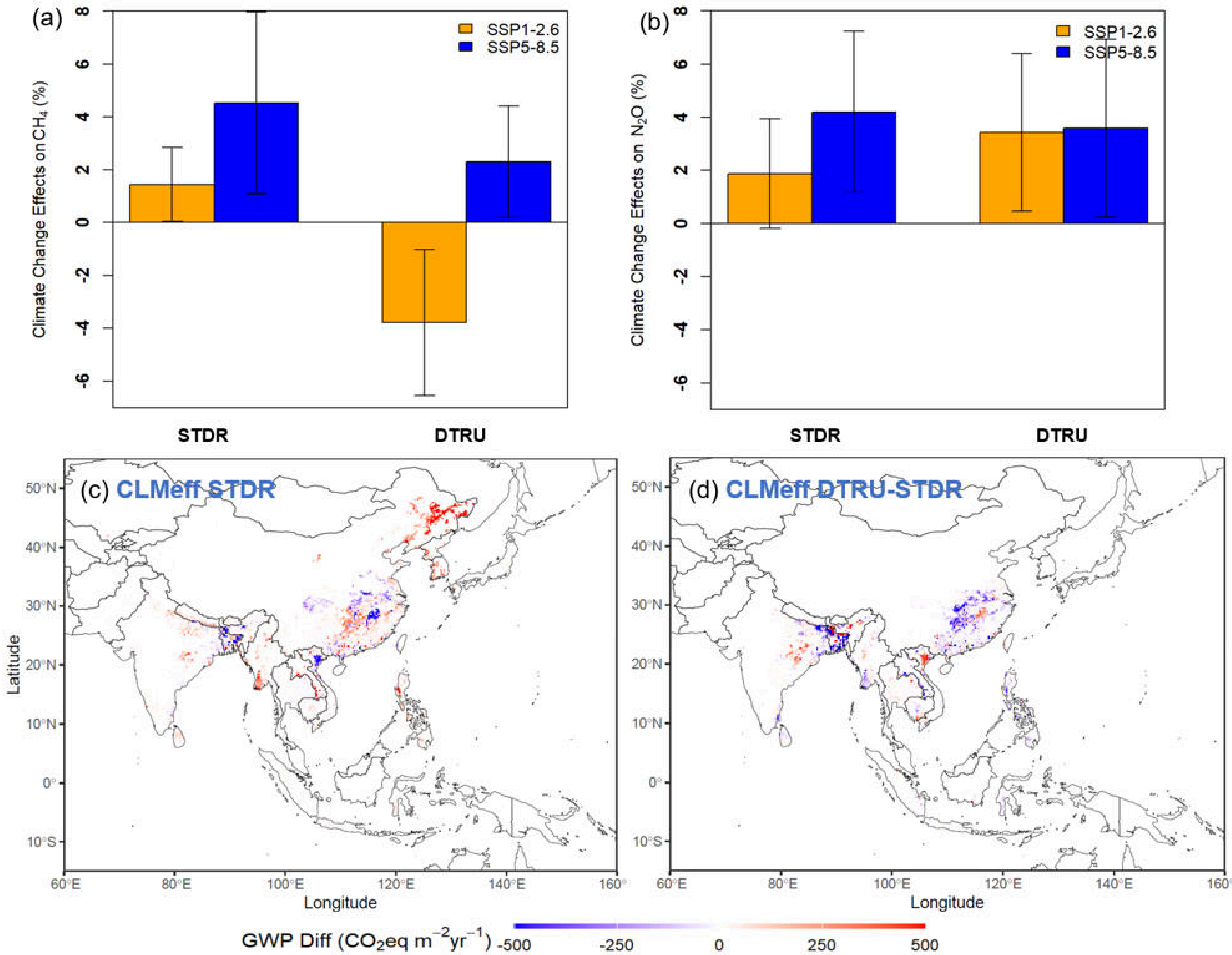

Figure 4

Projected climate-change impacts on CH_4 and N_2O emissions under single to double-rice pathway (STDR) and double-rice to rice-upland crops pathway (DTRU).

(a,b) climate change effects on CH_4 and N_2O emissions (mean ± 0.5sd), respectively; (c) spatial variations of climate change induced changes in GHG emissions with STDR pathway under SSP5-8.5; (d) spatial distribution of the difference in climate-driven GWP change with DTRU pathway relative to that of STDR under SSP5-8.5.

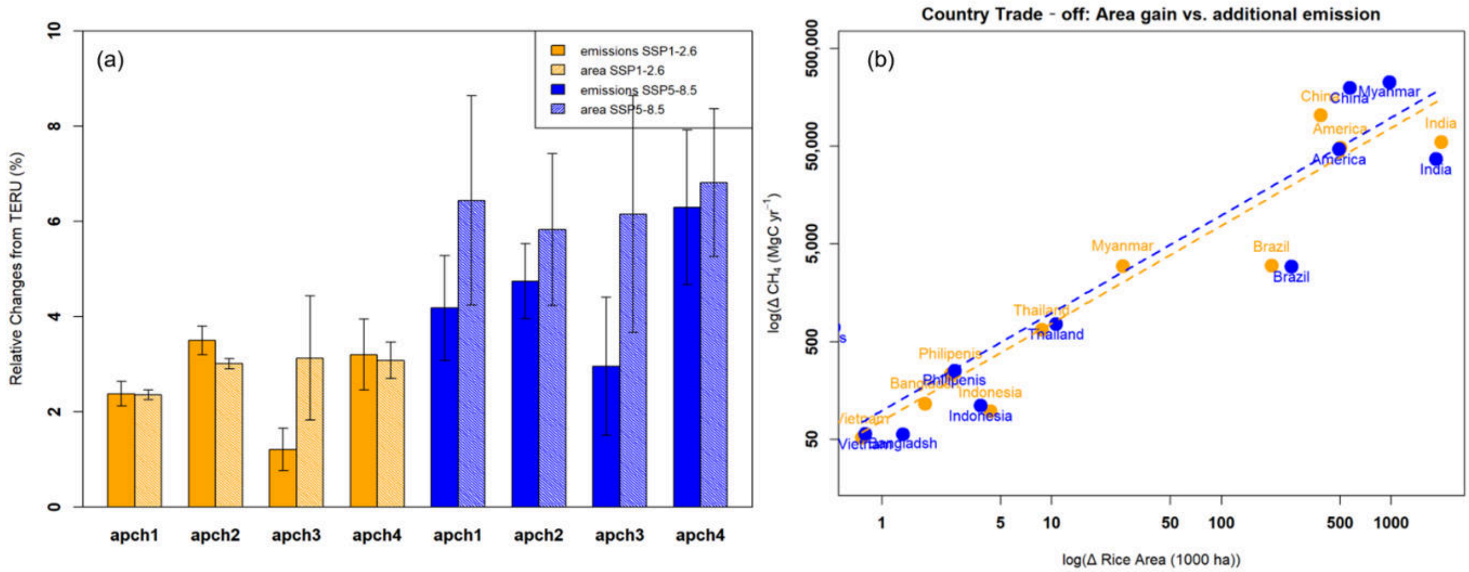

Figure 5

Impacts of targeted expansion of rice-upland crops systems (TERU) on CH_4 emissions and harvested rice area following the implementation of DTRU pathway.

(a) Mean relative change in GHG emissions and harvested rice area globally for four TERU approaches (apch1–apch4; see Table S2) under SSP1–2.6 and SSP5–8.5 scenarios. Error bars denote ± 0.5sd. (b) Relationship between national-scale changes in CH_4 emissions and rice-harvested area. Points represent individual countries (labels adjacent to markers); dashed lines show the global mean emission rates, below which indicates more gain in rice-cultivation area with less additional GHG emissions. All data were log transformed for visualization.

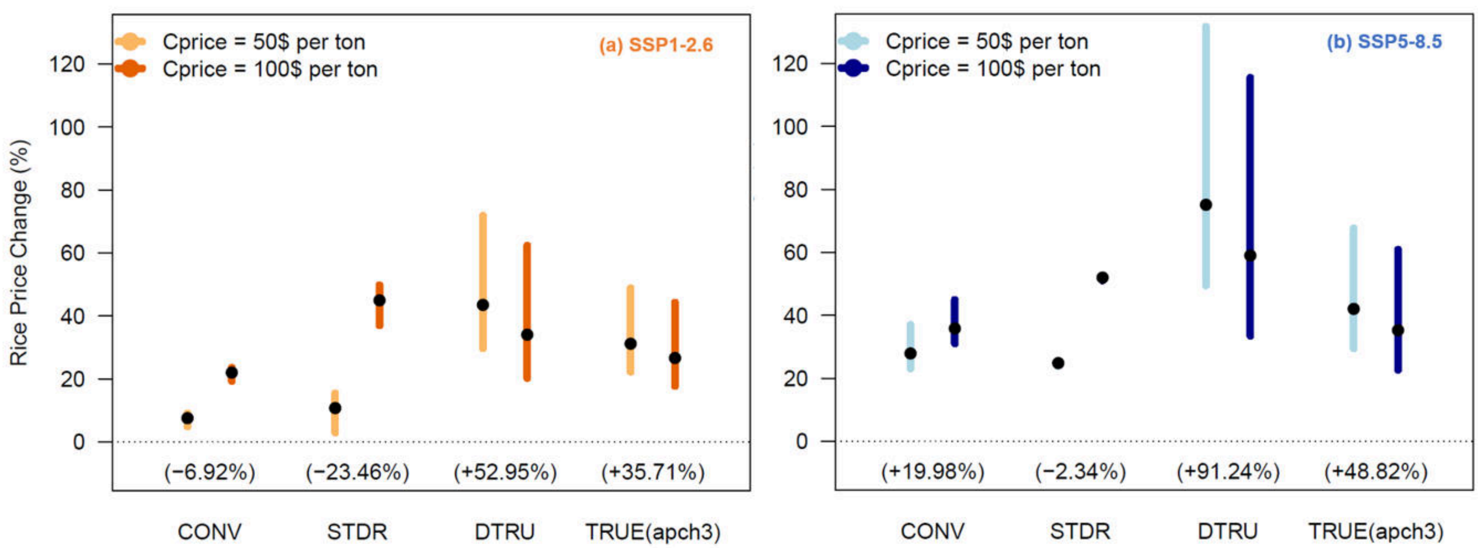

Figure 6

Global mean projected rice price changes (2050 – 2075 vs 1995 – 2014) under different rice-based cropping patterns.

The ranges reflect the uncertainty from varying constant of demand and supply elasticities (see Method section). The black dots denote the optimal projection. Bars at the base of each pathway show percent of price changes before applying carbon taxes/credits, i.e., the market-only balance of supply and demand.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- supplementmaterialsv4.docx
- Extendeddata.docx